



Wenyu Chiou

Ph.D. Candidate · Applied AI & LLM-Agent Systems

Building governed LLM-agent systems for human–environment interaction, grounded in empirical data and interpretable rule-based models.
Focus areas: LLM agents, human–environment interaction, agent governance, behavioral simulation, multi-agent systems, and AI research workflows.

✉ wec324@lehigh.edu +1 (484) 937-2528 Bethlehem, PA, USA github.com/WenyuChiou

23+

Open-source repos
shipped on GitHub

6+

LLM-agent systems
built & maintained

15

AI research skills
Claude Code skill ecosystem

1.8K+

GitHub stars
awesome-agentic-ai-zh

4

Research Manuscripts &
Articles
+ 4 conference talks

EXPERIENCE

Ph.D. Candidate

Aug 2024 — Present

Complex Adaptive Water Systems (CAWS) Lab, Lehigh University

- Built a governed LLM-agent framework (WAGF) with validation, execution control, and audit logs — simulates human–environment feedback grounded in empirical survey data.
- Coupled LLM-driven multi-agent system with Catastrophe Flood Model (CAT) for household flood adaptation.

LLM-Driven Agent Multi-Agent System ABM CAT Flood Adaptation

Research Assistant

Aug 2023 — Jun 2024

Surface Hydrology Group, IHOS, National Central University

- Conducted 2D/3D groundwater numerical modeling and data calibration for the Taoyuan coastal zone using HydroGeoChem (HGC).
- Evaluated Nature-Based Solutions (NBS) for groundwater and watershed management, assessing performance and hydrological impacts.

Groundwater Hydrology Numerical Simulation Geophysical Methods Groundwater Monitoring

OPEN-SOURCE · LLM AGENT SYSTEMS & AI RESEARCH WORKFLOWS

★ FLAGSHIP Multi-Agent LLM Governance · 2026

WAGF — Water Agent Governance Framework

Governed LLM-agent framework · Human–environment interaction

Validates LLM-agent decisions with domain rules, behavioral theory checks, retry feedback, and audit logs before simulation state updates.

LLM-Driven Agent Multi-Agent System Coupling Framework github.com/WenyuChiou/WAGF

ai-research-skills

AI research design & workflow skills · Apr 2026

Supports literature review, gap identification, feasibility scoping, manuscript writing, and cross-agent delegation.

Claude Code Plugin Research ai-research-skills

awesome-agentic-ai-zh

★ 1.8K+

Community · Education · Apr 2026

Trilingual learning map with 240+ curated resources from LLM basics to multi-agent systems.

Trilingual Community awesome-agentic-ai-zh

agent-collab-skills

Multi-Agent System · Orchestration · Apr 2026

Solves token blow-up and silent drift in multi-agent runs — 6–7× avg. token saving (17–22× peak).

Multi-Agent Workflow Claude Code Codex Gemini agent-collab-skills

codex-delegate

AI Research Workflow Tooling · 2025

Routes heavy code to Codex; reasoning stays in Claude — cost-aware multi-model routing.

Claude Code Codex CLI Multi-Agent Workflow codex-delegate

RESEARCH ARTICLES & MANUSCRIPTS

- Chiou, et al. (2026). Household flood adaptation and financial outcomes for homeowners and renters: A coupled agent-based and catastrophe flood modeling framework. Under review · *Journal of Hydrology*
- Yang, Y. C. E., & Chiou, W. Y. (2026). Leveraging large language models for agent-based simulation of human–water system interactions. Under review · *Water Resources Research*
- Chiou, et al. (2026). Divergent pathways in flood adaptation: Differences in decision-making factors and perceptions across socially disadvantaged populations. Under review · *Sustainable Cities and Society*
- Chiou, et al. (2021). Seasonal variations of water and energy budget of evergreen broad-leaved forest in central Taiwan. *Journal of Taiwan Agricultural Engineering*, 67(4), 62–75. (Mandarin with English abstract)

RESEARCH AREAS

INTERESTS & FOCUS

LLM Multi-Agent System Agent-Based Modeling
Catastrophe Modeling Human-Water Interaction
Decision-Making Process Structural Equation Modeling

CORE STACK

AGENT SYSTEMS

LLM agents Multi-agent systems Structured prompting
Validation rules Audit logs Simulation coupling

LANGUAGES

Python TypeScript R MATLAB C++

SOFTWARE

Git Jupyter SPSS AMOS ArcGIS Pro

AI TOOLING

Claude Code Codex CLI Gemini CLI Cursor
Ollama MCP Plugin / Skill

EDUCATION

Ph.D.

Lehigh University, United States

Dissertation: Advancing Coupled Human-Flood Modeling: From Empirical Foundations to LLM-Driven Multi-Agent Simulation
Aug 2024 — Present

M.S.

National Central University (NCU), Taiwan

Thesis: Investigating groundwater flow characteristics of the Taoyuan coastal zone.
Sep 2021 — Jun 2023

B.S.

National Central University (NCU), Taiwan

Sep 2017 — Jun 2021

LANGUAGES

Mandarin Chinese · Native

English · Professional

CONFERENCE PRESENTATIONS

Household Flood Adaptation and Financial Outcomes under Social Heterogeneity: A Coupled Agent-Based and Catastrophe Modeling Framework.
CERCat Semi-Annual Update · 2026

Modeling Long-Term Flood Adaptation Action Across Marginalized and Non-Marginalized Groups: A Coupled Agent-Based Model.
AGU Fall Meeting 2025 · 2025

Investigating the Decision-Making Process and Causal Relationships between Flood Risk Perception and Adaptation Actions.
2nd International Sociohydrology Conference · 2025

Integrating ERT, Field Observations, and Numerical Simulations to Investigate SGD Dynamics of the Taoyuan Tableland, Taiwan.
AGU Fall Meeting 2023 · 2023